Lab experiment-6

Vacuum Cleaner Problem

Aim

To write a Python program to simulate the Vacuum Cleaner Problem.

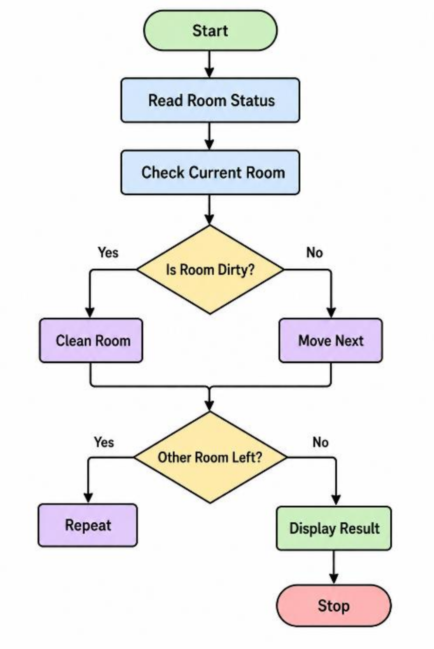
Objective

To clean all dirty rooms using a simple vacuum cleaner agent.

Algorithm

1. Start.
2. Read the status of Room A and Room B.
3. Place the vacuum in the starting room.
4. If the current room is dirty, clean it.
5. Move to the other room and repeat.
6. Display the final status.
7. Stop.

Flowchart

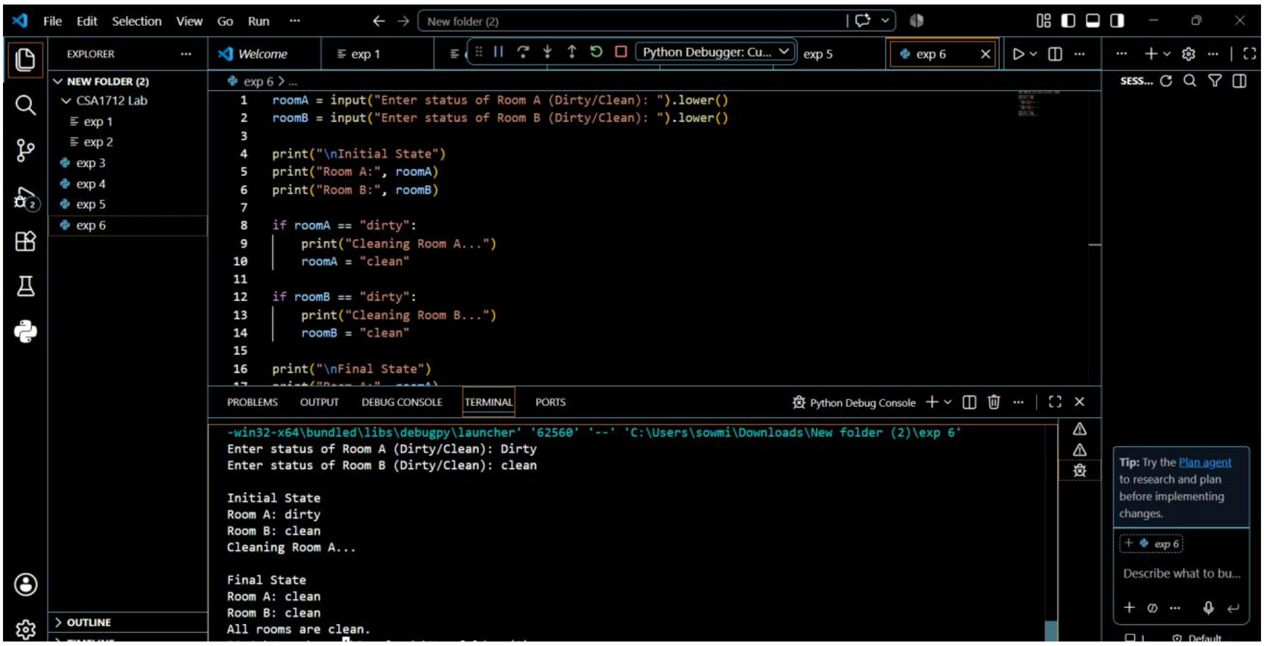


Code

```
roomA = input("Enter status of Room A (Dirty/Clean): ").lower()
roomB = input("Enter status of Room B (Dirty/Clean): ").lower()

print("\nInitial State")
print("Room A:", roomA)
print("Room B:", roomB)

if roomA == "dirty":
    print("Cleaning Room A...")
    roomA = "clean"
```

```
if roomB == "dirty":
    print("Cleaning Room B...")
    roomB = "clean"


print("\nFinal State")
print("Room A:", roomA)
print("Room B:", roomB)
print("All rooms are clean.")
```

Output



Result

The vacuum cleaner agent cleaned all dirty rooms successfully.

Conclusion

The program successfully simulates the Vacuum Cleaner Problem using a simple intelligent agent.